

1. Dataset Overview

The dataset used in this project is the **Turkish Cities Daily Weather Dataset**, obtained from Kaggle:

Name: Turkish Cities Daily Weather Dataset

Source: Kaggle (Contributor: Bahadır Bor)

URL: <https://www.kaggle.com/datasets/bahadirbor/turkish-cities-daily-weather-dataset>

This data consists of daily weather observations at many cities of Turkey. The program helps predict, analyze, and monitor the weather and climate at the city level. The continuous temporal and spatial coverage of the dataset makes it suitable for agricultural and environmental studies.

2. Motivation for Dataset Selection

The data was chosen for the following reasons.

Geographical Relevance.

The system is aimed at farmers in Turkey, therefore the data used must be local weather data. This dataset is only with Turkish cities, meaning that its data source and application domain are geo-aligned.

Daily Temporal Resolution.

The project focuses on providing decision support for short-term (1–3 day) time frames which is strongly affected by weather dynamics. The dataset's daily granularity is appropriate for.

Frost risk detection.

Rainfall-based agricultural planning.

Irrigation and planting analysis.

Suitability for Time-Series Modeling.

The data from each city creates a time series which can be univariate or multivariate.

Lag-based feature engineering.

Trend and seasonality analysis.

Baseline and machine learning forecasting experiments.

Practicality for Academic Projects.

The dataset can be publicly obtained while it has a manageable format such as CSV and is easily processed using standard data science software (e.g. Python, Pandas). Furthermore, it is suitable for a graduation project as it does not require a lot of time or effort.

3. Dataset Structure and Attributes

The dataset consists of daily records, where each row corresponds to **one city on one specific date**. Typical attributes include:

Attribute	Description
city	Name of the Turkish city
date	Observation date (daily frequency)
temperature	Daily average temperature (°C)
temp_min (<i>if available</i>)	Daily minimum temperature (°C)
temp_max (<i>if available</i>)	Daily maximum temperature (°C)
precipitation	Daily precipitation amount (mm)
humidity (<i>optional</i>)	Relative humidity (%)
pressure (<i>optional</i>)	Atmospheric pressure (hPa)

The exact set of columns may vary slightly depending on the dataset version. All temperature values are expressed in degrees Celsius, and precipitation is measured in millimeters.

4. Temporal and Spatial Coverage

Data should cover multiple years in the range of 3 - 5 years depending on the city.

Dozens of cities from different climate zones in Turkey including;

Coastal regions.

Continental inland areas.

High-altitude cities.

Due to this diversity, different regions can be compared and generalized agri decision rules can be developed.

5.Data Preprocessing and Preparation

Prior to the application of the dataset, we applied the following preprocessing step.

Datetime Parsing.

The date column was converted into a datetime format to enable time series analysis.

Missing Value Handling.

Short gaps were filled using forward or backward filling.

Data rows with too many NA values were removed.

Feature Engineering.

Additional features were derived, including.

Previous days' temperatures and precipitations are lagged values.

Rolling averages (3-day and 7-day windows).

Seasonal indicators (month, day of year).

Rainfall Labeling.

For classification tasks, binary rainfall indicator was created.

```
rain_label = 1 if precipitation > 0.1 mm  
            0 otherwise
```

6. Role of the Dataset in the Project

The data set generates additional insight and analysis but is not the primary operational forecast. Specifically, it is used for.

Historical Climate Analysis.

Learning about temperature and rainfall patterns in Turkish cities for agricultural thresholds.

Baseline Model Development.

Putting in place models like persistence ones, regressive or classification for comparison

Model Evaluation and Benchmarking.

We will compare our machine learning model results against baseline methods to see how they performed.

Validation of Agricultural Decision Rules.

We Are Testing Frost Alerts Irrigation Rules Planting Suitability Logic Against Actual Historical Data.

7. Dataset Limitations

The dataset has a lot of limitations even if useful.

It only provides historical observations not future forecasts.

The dataset is daily, so it does not show intra-day variations like wind speed every hour.

Does not incorporate parameters unique to crops or soils.

Some cities may have incomplete records for certain years.

Because of these limitations, a real-time weather API is used in the functioning system to create real 1–3 day forecasts, while this dataset is used for evaluation, analysis and academic justification.

8. Summary

In conclusion, this dataset enables real-life data collection to analyze historical data and set a specific base for modeling in the context of the project. When we add a real-time weather API, we will have a powerful framework suitable for scientific assessment and agricultural decision-making.